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Full Length Article

Evolution of cooperation in spatial heterogeneous social dilemmas with dynamic self-adaptive matching

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ABSTRACT

From the perspective of real-world interaction scenarios, the assumption of monotypic social dilemma engagement or fixed role allocations within heterogeneous social dilemmas contexts fundamentally misrepresents the dynamics of human social cognition. Empirical evidence consistently suggests that behavioral plasticity is achieved through adaptive identity switching across heterogeneous social dilemma environments. Herein, we develop an evolutionary game model simultaneously encompassing multiple social dilemmas (prisoner's dilemma game and snowdrift game), wherein agents engage in polymorphic game-environment governed by their own adaptive characteristic. Specifically, during the strategy update phase, if the agent $x's$ current strategy is retained, its strategy stickiness θ_x increases one unit; otherwise, the metric reverts to the initial value of 0. Additionally, if θ_x reaches the maximum value of 100, it signifies the termination of the old agent and the initialization of a new one, with the metric also reverting to 0. Moreover, agent whose strategy adherence is beyond predefined threshold θ_{th} is classified as possessing strong social dilemma resolution capacity, while that is below the threshold is allocated to mild-conflict scenarios. Subsequently, through sufficient Monte Carlo simulation, we systematically investigate the evolutionary trajectories and stable-state characteristics of cooperation in dynamically adaptive multi-social dilemma systems, yielding several insightful findings.

1. Introduction

As is widely recognized, cooperation characterized by pro-social attributes can strengthen social resilience through optimizing resource allocation and alleviating social conflicts. [1–3]. However, cooperation with altruism requiring sacrifices theoretically fails to achieve evolutionary stability in selfish population [4–7]. Therefore, elucidating the underlying mechanisms behind the emergence and maintenance of cooperation has become a long-standing scientific challenge in the study of complex systems. It forms the cor-

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nerstone of foundational research in theoretical biology, computational science, social science, and nonlinear dynamics, attracting significant attention from scholars across diverse fields [8–13].

Up to now, a series of game theory models involving conflict scenarios have been proposed to objectively de-construct the internal reasons for the evolution of cooperation, including prisoner's dilemma game (PDG) [14,15] and snowdrift game (SDG) [16,17]. In general, agent who contributes resources to generate positive externalities are defined as cooperator (C), whereas the one who appropriates benefits through non-reciprocal exploitation are categorized as defector (D). Within these above dyadic game frameworks, bilateral cooperation yields the reward R, whereas mutual defection results in the punishment P. Under asymmetric conditions, however, the cooperator incurs the sucker's payoff S, while the defector accrues the temptation to defect T through opportunistic exploitation. The critical distinction between canonical formulations of the PDG and SDG manifests in the divergent ranking hierarchies of their respective payoff parameters ($T > R > P > S$ vs. $T > R > S > P$), which is formalized in the strategic interaction paradigm of game theory. This parametric divergence engenders diametrically opposed solutions in their respective strategic formulations, with the former exhibiting the dominant-strategy equilibrium (D,D) [18] and the latter demonstrating the coexistence equilibrium (D,C) or (C,D) [19] under standard replicator dynamics [20]. Therefore, the PDG reveals the core paradox of cooperation, which arises from structurally embedded incentive incompatibilities between individual rationality and collective welfare maximization. While the SDG indicates attenuated incentive conflicts conducive to conditional cooperation under evolutionary stable strategies.

At the same time, considered the paradoxical divergence between cooperative behavioral outcomes and orthodox full rational choice paradigms, evolutionary game theory (EGT) has emerged as a fundamental analytical paradigm for decoding the dynamics of cooperation. It systematically incorporates the principle of bounded rationality, which manifests in strategic decision-making processes under incomplete information into the modeling framework of cooperative behavior [21]. Under this theoretic paradigm, Nowak has summarized five laws of governing cooperation [22], such as kin selection [23], group selection [24], direct [25] and indirect [26] reciprocity, and network or spatial reciprocity [27,28]. The above mechanisms collectively constitute the foundational pillars of cooperation emergence under natural selections. Nonetheless, on the one hand, it is more relevant to study the emergence of cooperation in populations composed of unrelated agents. On the other hand, the ability of any agent to take part in game is limited, making that they can only be able to interact in a limited neighborhood. Thus, network or spatial reciprocity have garnered significant attention [27], in which cooperators effectively resist the endless attacks of defectors by forming close clusters. The focus enables researchers, within the EGT framework, to explore group interactions involving larger participants extended time scales, thereby shedding light on the causes of the ubiquitous phenomena of cooperation. Up to now, a large number of mechanisms have been shown to significantly govern the evolutionary persistence of cooperation, including but not limited to mobility [29,30], punishment [31], memory [29,32], and payoff control [33,34]. In particular, during this process, many intriguing phenomena were inadvertently discovered, such as dynamic cyclic dominance [35–37] and role polarization [38], which has significantly enriched and advanced the theory of the evolution of cooperation.

It should be pointed that, most of theoretical models usually assumed that there was a single social dilemma in the system, and then carried out the research of the evolution of cooperation. Therefore, such simple assumption may inadvertently conceal the heterogeneity of agents and their environmental perception, thereby limiting the generalization of the findings to the complex real world. In reality, due to differences in plasticity phenotypes, kinds of agents in complex systems may have different cognition of the same issue, which leads to extremely complex social dilemmas in the real system. Hence, while traditional game theory prioritized single-game analyses, recent explorations of the evolution of cooperation have achieved a paradigm shift toward multi-game dynamics. Cressman et al. [39] initially developed a multi-game dynamics framework through two independent decision scenarios. Their results showed that the system's equilibrium state could be effectively characterized by each game dynamics. Multi-game theory introduces an innovative modeling approach where heterogeneous payoff matrices operate concurrently within the single evolutionary competition framework. Then, Hashimoto [40] formulated a replication dynamic equation with multi-game, where the total payoff for agent was a simple sum of the payoff for each independent game. The research proved through theoretical analysis and numerical simulation that multi-game dynamics cannot be simply decomposed into the dynamics of every single game. In other words, the emergent complexity and intrinsic unpredictability characterizing multi-game dynamics constitute fundamentally irreducible properties absent in single-game environments. Furthermore, Hashimoto [41] also systematically studied the dynamic behavior of multi-game effects in finite populations for the first time. By incorporating a stochastic process characteristic, he elucidated the distinctive behavior of multi-game effects in finite populations, particularly the correlations between strategies.

Given the fundamental differences in evolutionarily stable solutions between structured populations and well-mixed environments, Wang et al. [42] extended multi-games including both PDG and SDG into lattice network, significantly enhancing its practical relevance. The outcomes indicated that payoff heterogeneity spontaneously gave rise to cooperative leadership nuclei, which established directed strategy transfer dynamics with an inherent bias toward cooperative phenotypes through evolutionary selection pressures. On the contrary, through innovative parametrization of temptation payoff gradients, Szolnoki and Perc [43] proposed a breakthrough multi-game model only integrating the prisoner's dilemma game. This novel architecture enables dynamic inversion of invasion thresholds between cooperation and defection generates self-sustaining cooperative propagation through evolutionary feedback loops. Furthermore, Szolnoki and Chen [44] expanded the interaction mode among individuals from the single cooperation and defection to the combination of multiple strategies including pair-point (prisoner's dilemma game) interaction and multi-point (public goods game) interaction. This extension enabled the model to more closely reflect the diverse choices of individuals in social interactions in the real world, and revealed complex dynamic behaviors and counterintuitive phenomena. Since then, the methodological advancement in spatial multi-game theory has not only gained substantial academic recognition, but has also catalyzed interdisciplinary synthesis across multiple research frontiers. For instance, Chowdhury et al. [45] integrated the ecological construct of *free space* with evolutionary game theory, constructing an ecological-evolutionary framework that incorporates punishment, co-

operation, and defection strategies through the synergistic combination of PDG and SDG architectures. Han et al. [46] constructed a spatial multi-game model combining age factors, in which agents were assigned to participate in either PDG or SDG by comparing the age threshold θ with the value of agent's age. However, it must be noted that, the age of the agent does not dynamically change over time once it set. Moreover, extending spatial multi-games to the two-layer coupled network can allow for the design of complex social structures that more closely resemble real-world conditions. To this end, Jiang et al. [47] constructed a two-layer network model coupling the scale-free network and the regular lattice network. In this model, participants on the scale-free network engaged in the prisoner's dilemma game, while those on the regular lattice network took part in the public goods game. Within this multi-game environment, participants interacted with each other while being constrained by incentive mechanisms. The results indicated that through appropriate incentive mechanisms and network structure design, cooperative behaviors could be effectively promoted and wealth inequality might be reduced.

Unfortunately, however, the current research generally assumes that agent participates in one of types of the multi-games with the given identity and does not change during the evolution process. Obviously, this simple assumption does not correspond to the actual situation. After all, it ignores the exact match between the agent's dynamic adaptability and the corresponding social dilemmas. For instance, the human life cycle follows a developmental trajectory wherein childhood maturation progresses through adulthood, ultimately yielding to generational succession as aging populations are supplanted by new birth cohorts. Meanwhile, in reality there are significant differences between children and adults in the cognitive ability and behavioral willingness of social dilemmas. Analogous dynamics can also be systematically observed from corporate to national. Thus, we try our best to construct a novel multi-games dynamics model, where the dynamic evolution of agent identity dominates its ability to participate in the corresponding social dilemma, ensuring the homeomorphic mapping between the identity spaces and the social dilemma spaces. Operationally, although we also assume that there are both PDG and SDG types of social dilemmas in the system, what kind of social dilemmas the agent can perceive is determined by the identity.

Admittedly, there are numerous factors in real life that constrain strategic or identity changes in humans or other entities. However, for the purpose of accentuating the influence of such dynamic transformations on the evolutionary trajectory of cooperation, we have opted for a simplified modeling approach. Herein, we quantify the agent $x's$ ability to maintain their current strategies, which is defined as strategy persistence or stickiness θ_x. The strategy persistence or stickiness dynamically evolves according to the established protocol rules, which will be elaborated later in detail. Then, by comparing the strategy persistence θ_x of agent x with the critical threshold θ_{th}, we simply distinguish agents into two sub-groups. To be specific, if the agent $x's$ strategy persistence θ_x do not exceeds the threshold θ_{th}, he will be categorized into the juvenile sub-group; otherwise, he can be divided into the adult sub-group. From the socio-structural perspective, juvenile populations predominantly occupy the role of recipients within protective social provisioning systems, whereas adult cohorts institutionally designated as primary bearers of institutionalized role obligations. Thus, agent who is divided into adult sub-group must take part in PDG, whereas agent who is categorized into juvenile sub-group can participate in SDG. It should be pointed out that, the agent's strategy persistence exhibits functional isomorphism with age-dependent inertia mechanisms in evolutionary frameworks. Nevertheless, our age characteristic evolves dynamically with the interaction process, which is fundamentally different from the static age settings in previous studies [48,49]. Moreover, it is pointed out that our model achieves dynamic alignment between agent identities and social dilemmas, enabling precise characterization of multi-games dynamics.

The remainder of this paper is systematically structured as follows. First of all, the dynamic self-adaptive heterogeneous social dilemmas model is comprehensively elaborated in Section 2. Subsequently, we present the multidimensional simulation analysis in Section 3. Lastly, we synthesize the principal findings and contributions in Section 4.

2. Model

2.1. Network structures

To underscore the critical impact of adaptive identity reconfiguration in agents and the contextual alignment of social dilemmas on the evolution of cooperation, we employ the $N = L \times L$ square lattice network with the boundary conditions and Von Neumann neighborhood (*i.e.* $< k > = 4$) as the foundational interaction topology. It must be pointed out that, the node degree distributions in real word networks predominantly conform to either Poissonian statistics or power-law distributions, and the diameter of some networks may also exhibit small-world characteristics in numerous empirically observed systems. Therefore, it is necessary to conduct topological sensitivity analysis within the established framework to quantify robustness metrics across structural perturbations. In this paper, the methodological encompasses six other network architectures: Erdös-Rényi random network, Watts-Strogatz small-world network, Barabási-Albert scale-free network, regular small-world network, random regular network and square lattice network with Moore neighborhood.

To clarify the generation patterns of these different types of network structures, it is necessary to provide a brief explanation. For Erdös-Rényi random network, Watts-Strogatz small-world network, regular small-world network and random regular network, all these networks are generated by rewiring the edges of specific networks. The differences lie in the varying probabilities of edge rewiring p and whether the degree of nodes remains unchanged. By rewiring the edges in nearest-neighbor regular network with probabilities $p = 1$ and $p = 0.03$, we generate Erdös-Rényi random network and Watts-Strogatz small-world network, respectively. According to the previous work [50], by rewiring the edges in square lattice network with probabilities $p = 0.03$ and $p \to 1$, we obtain regular small-world network, random regular network, respectively. It is crucial to emphasize that, during the process of rewiring edges in specific networks, self-connections and multiple connections between the same pair of nodes are not permitted. In addition, during the process of reconnecting broken edges, the degree distributions of the Erdös-Rényi random network and the Watts-Strogatz

small-world network gradually show strong heterogeneity. However, the degree distributions of regular small-world network and random regular network remain unchanged all the time. The Barabási-Albert scale-free network is generated through a preferential attachment mechanism. To be specific, we can first construct a fully connected network with m_0 nodes. Then, we add a new node every time and connect each new node to the m ($m < m_0$) existing nodes in the current network with probabilities proportional to their degrees. After performing the aforementioned operation t times, we will obtain a scale-free network with $N = m_0 + t$ nodes and $M = mt + M_0$ edges, where M_0 represents the initial number of edges in the network. In this study, we set $m_0 = 3$ and $M_0 = 3$. If we also connect the diagonal nodes in the square lattice network with Von Neumann neighborhood, we will gain the square lattice network with Moore neighborhood.

Moreover, the degree distributions of Erdös-Rényi random network, Watts-Strogatz small-world network and Barabási-Albert scale-free network exhibit the high degree of heterogeneity. What is more, the degree distributions of Erdös-Rényi random network and Watts-Strogatz small-world network follow the Poisson distribution, whereas the degree distribution of Barabási-Albert scale-free network follows the power-law distribution. The degree distributions of the regular small-world network, random regular network, and square lattice network with Moore neighbors show the high degree of homogeneity. Except for that the average degree of square lattice network with Moor neighborhood is $< k >= 8$, the one of other networks is $< k >= 4$.

2.2. Initial setup

During the initialization phase, every agent residing on the network point must stochastically make cooperative ($s_x = (1,0)^T$) or defective ($s_x = (0,1)^T$) strategies with equiprobability under complete informational opacity regarding. Complementing this foundation, we introduce a strategic inertia parameter (denoted as θ_x) to characterize agent $x's$ ability for strategy conservation, formally quantifying the persistence magnitude of current strategy within the evolutionary framework. Furthermore, it should be noted that, although we set the initial value of θ_x to be 0, the simulation shows that the initial distribution of this value has no fundamental impact on the results. Simultaneously, we posit the existence of two distinct types of social dilemmas within the system: the prisoner's dilemma game (PDG) and the snowdrift game (SDG). Since agents will take part in the game around the focal agent, the type of game is determined by the focal agent x. In this paper, the payoff matrix of PDG is,

$$M = \begin{pmatrix} 1 & -r \\ 1+r & 0 \end{pmatrix} \tag{1}$$

In addition, the payoff matrix of SDG adheres to a comparable configuration,

$$M = \begin{pmatrix} 1 & 1-r \\ 1+r & 0 \end{pmatrix} \tag{2}$$

where $0 < r < 1$ is the cost-to-benefit ratio, which allows us to quantify the social dilemmas intensity of PDG and SDG within a unified framework. Furthermore, what kind of game the agent participates in depends on the identity determined by the strategy stickiness, which will be described later in detail.

2.3. Payoff calculation and strategy update

Following standard Monte Carlo procedure, an agent x in pairwise game is stochastically activated to sequentially play games with its adjacent neighbors. The accumulation process of the payoff is formalized as,

$$P_x = \sum_{y \in \Omega_x} s_x^T M s_y \tag{3}$$

where Ω_x indicates the size of $x's$ neighborhood.

Then, the focal agent x asynchronously update its strategy. To be specific, any neighbor y of the agent x is chosen at random and gains the payoff P_y in the similar way. Afterwards, the strategy update process follows pairwise payoff comparison between the focal agent x and the randomly sampled neighbor y, governed by the Fermi-Dirac distribution as [51],

$$W_{(s_x \leftarrow s_y)} = \frac{1}{1 + \exp[(P_x - P_y)/K]} \tag{4}$$

where K controls agent's cognitive biases during social learning processes, reflecting its deviation from perfect rationality in evolutionary game dynamics. Without loss of generality and in alignment with established theoretical frameworks [53–55], we assign $K = 0.1$ throughout this study. In addition, to validate the model's generalizability and resilience, we also take into synchronous strategy update protocol across diverse parameter configurations.

2.4. Strategy stickiness protocol

Herein, we define the agent's ability to adhere to its current strategy as strategy stickiness and draw an analogy with the natural age characteristic. Specifically speaking, during the strategy update process, if the agent x keeps the current strategy, then the value of strategy stickiness θ_x increases one unit; otherwise, the value of θ_x is reset to the initial value 0, meaning the birth of a newborn. At the same time, to better capture the intrinsic aging dynamics, we impose a bounded rationality constraint by capping strategy

persistence at the maximum value of $\theta_{max} = 100$. In other words, the strategy stickiness can be only evolved within the interval [0,100]. When an agent's accumulated strategy stickiness θ_x exceeds the maximum, the system triggers simultaneous senescence and renewal events. More specifically, incumbent agent termination is algorithmically coupled with nascent agent initiation, which not only ensures that the size of population does not change but also preserves topological connectivity.

Moreover, since we here analogize strategy stickiness or persistence to natural age, this necessitates that we do not allow the maximum value of θ_{max} to be set excessively too large or too small when constructing our model. Meanwhile, our primary focus is on the threshold value, θ_{th}, that distinguishes between juveniles and adults in this study. Building upon the methodological framework established in prior studies [56,57], we use a dichotomous state variable $\zeta_x \in \{0, 1\}$ to track strategic volatility at the agent level. The ζ_x codifies strategy retention (1) or revision (0) events, which is consistent with discrete choice theory and can be record as,

$$\theta_x = \begin{cases} \theta_x + 1, & \zeta_x = 1 \\ 0, & \zeta_x = 0 \end{cases} \tag{5}$$

Then, through comparing agent $x's$ strategy stickiness θ_x with the predetermined critical threshold θ_{th}, we operationalize a socio-computational taxonomy that dichotomizes the population into juvenile populations ($\theta_x < \theta_{th}$) and adult cohorts ($\theta_x \geq \theta_{th}$). In general, juvenile populations, designated as the protected object, will engage in SDG characterized by moderate-intensity social dilemmas; conversely, adult cohorts are assumed the role of obligated stakeholders, mandatorily participating in PDG emblematic of high-stakes collective action problems. The agent's identity dynamically evolves after comparing its strategy stickiness θ_x with critical threshold θ_{th}, which effectively encodes age-structured social roles within evolutionary computation frameworks.

It should be specifically noted that by comparing θ_x with θ_{th}, distinguishing agents into two different types, namely the juvenile sub-group and the adult sub-group, does indeed simplify the model setup, but it is not without basis. The phenomena of categorizing groups by setting simple thresholds are ubiquitous in real life. For instance, for individuals, the law defines them as juveniles or adults by setting the age threshold. Similarly, for enterprises, they are differentiated into above-scale enterprises and regular enterprises based on a pre-set threshold for their main business income. Here, we employ this simple setup to capture the dynamic trajectory of cooperative evolution and explore the mechanisms underlying the emergence of cooperation. Meanwhile, despite the thresholds being predetermined, the strategic stickiness of agents undergoes dynamic evolution, which renders it difficult for us to accurately estimate and forecast the evolutionary dynamics of cooperation. Therefore, we will primarily use the Monte Carlo simulation to investigate the evolutionary trajectories of cooperation in heterogeneous social dilemmas with dynamic adaptive matching.

The scale of the square lattice network is $L = 100$, that is, $N = 10000$. Furthermore, we also consider some larger ($L = 200$, 300, and 400) and smaller ($L = 80$ and 50) networks, and obtain qualitatively consistent results. In the context of topological changes, the size of the network and the average degree $< k >= 4$ remain constant except for that the square lattice with Moore neighbors is $< k >= 8$. Within every full Monte Carlo step, each agent on average has a chance to update its strategy. It should be emphasized that the predetermined critical threshold parameter θ_{th}, which distinguishes agents as juveniles and adults, is a key factor affecting the evolutionary dynamics of cooperation in heterogeneous social dilemmas with dynamic adaptive matching. Therefore, we regard θ_{th} as the primary research subject. Meanwhile, the frequency of cooperation, F_c, serves as a critical metric for characterizing evolutionary dynamics, which is computed as the average over the final 5×10^3 of each 5×10^4 Monte Carlo simulation steps. Moreover, to mitigate stochastic fluctuations, we perform at least 20 independent realizations under identical conditions, ensuring the robustness of our findings.

3. Results

In this section, we first conduct a macroscopic exploration of the trends in the evolution of cooperation and then fit and compare the simulation results with the theoretical results obtained through the generalized mean-field approach. Next, we de-construct the microscopic mechanisms underlying the evolution of cooperation in heterogeneous social dilemmas with dynamic adaptive matching by analyzing time series and strategy snapshots, respectively. At last, we validate robustness and resilience through alternative network structures as well as strategy update pattern.

3.1. Preliminary macroscopic inquiry

In the main framework, we adopt square lattice network as the foundational topological configuration to establish nearest-neighbor game interaction model. Subsequently, the $F_c - \theta_{th}$ dependence is rigorously delineated through phase-space analysis.

As theoretically posited in our framework, the critical threshold θ_{th} stratifies agents into juvenile sub-population and adult sub-cohort. Distinct sub-population engages in divergent games, which fundamentally reshape cooperation-defection equilibria in spatial heterogeneous social dilemmas. Therefore, for different critical threshold θ_{th}, we plot the frequency of cooperation F_c as the function of the cost-to-benefit radio r from the macroscopic perspective, as shown in Fig. 1. Here, the cost-to-benefit radio r is the key parameter that regulates the intensity of social dilemmas. The greater the value of r, the more pronounced the strategic contradictions in the decision-making environment become. So it can be found that, overall, regardless of θ_{th}, the frequency of cooperation F_c manifests a monotonic nonlinear decrease as r increases. In particular, all agents in the system take part in PDG when $\theta_{th} = 0$. Extensive theoretical investigations have demonstrated that, PDG environments exhibit inherent dissipation characteristics, which destabilizes and destroys the formation of cooperation while accelerating defection-attractor dominance. Therefore, even for quite small value of r, cooperation quickly annihilates. However, when $\theta_{th} > 0$, there will always be a subset of agents whose strategies adherence

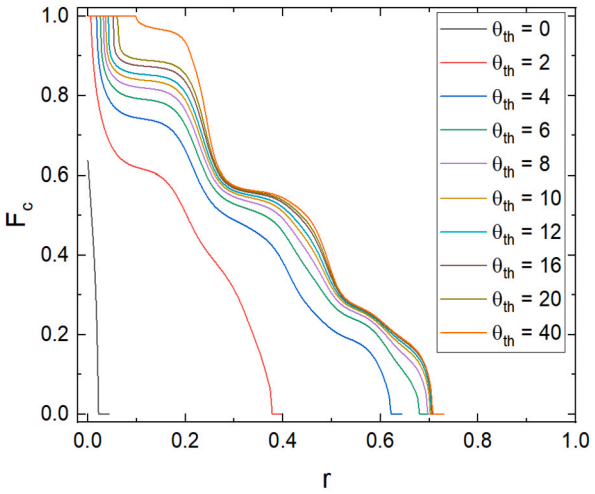


Fig. 1. Under different values of critical threshold θ_{th}, the frequency of cooperation F_c as functions of cost-to-benefit radio r at the equilibrium phase. Some of the main parameters are $L = 100$, $MCS = 5 \times 10^4$, $< k >= 4$, and $K = 0.1$.

fail to surpass the critical threshold θ_{th} are algorithmically categorized into juvenile cohorts, so that they take part in the SDG. In contrast to the PDG, the SDG exhibits less severe strategic contradictions between individual rationality and collective welfare, thereby fostering more favorable conditions for the evolution of cooperation. So, with the increase of θ_{th}, the system even appears to be full cooperation. Meanwhile, it can be found that, the threshold of cooperation annihilation r_c increases with the increase of θ_{th}. Notably, under substantial threshold magnitudes (e.g. $\theta_{th} = 40$), the extinction threshold for cooperation r_c is about 0.7, which becomes quantitatively indistinguishable from the pure SDG baseline. These findings reveal asymptotic convergence of evolutionary dynamics toward canonical SDG equilibrium configuration with the increase of critical threshold θ_{th}.

Furthermore, to verify the correctness of the numerical simulation results, it is necessary to calculate the theoretical frequency of cooperation F_c based on the generalized mean field theory. First, we define $W_{C\to D}$ to represent the probability that a cooperator transitions into a defector during the evolutionary process. Similarly, we also define $W_{D\to C}$ to indicate the probability that a defector transitions into a cooperator during the evolutionary process. It is important to note that under this update rule, the probability of each agent changing their current strategy is unrelated to the payoffs of others. Thus, $W_{C\to D}$ and $W_{D\to C}$ can be calculated as follows,

$$W_{C\to D} = \frac{1}{N_C} \sum_{s_i=C} W_i \tag{6}$$

$$W_{D\to C} = \frac{1}{N_D} \sum_{s_i=D} W_i \tag{7}$$

where N_C and N_D represent the number of cooperators and defectors in the system at the current moment, respectively. Meanwhile, $N_C + N_D = N$. In this context, we use the generalized mean-field theory to analytically determine the theoretical frequency of cooperation F_c. According to [52], the dynamic process of F_c can be expressed as,

$$\dot{F_C} = (1 - F_c)W_{D\to C} - F_c W_{C\to D} \tag{8}$$

When the system reaches a steady state, the value of $\dot{F_c}$ is 0. Therefore, the theoretical frequency of cooperation F_c can be calculated as,

$$F_C = \frac{W_{D\to C}}{W_{D\to C} + W_{C\to D}} = \frac{1}{1 + \frac{W_{C\to D}}{W_{D\to C}}} \tag{9}$$

Next, the theoretical frequency of cooperation F_c can be gained based on the above analysis. Meanwhile, to obtain the complete effect of the critical threshold θ_{th} on the evolution of cooperation, Fig. 2 presents the frequency of cooperation F_c as function of the critical threshold θ_{th} under kinds of the cost-to-benefit radio r. In particular, to enhance the credibility of the simulation results, we fit and compare them with the outcomes derived from the generalized mean-field approach. Notably, due to the large scale of the network, it is very difficult for us to directly calculate the values of $W_{C\to D}$ and $W_{D\to C}$ through theoretical analysis. Hence, these transfer probabilities used for theoretical analysis of the frequency of cooperation F_c are obtained indirectly through simulation. In Fig. 2, continuous dot-lines represent the simulation results, while discrete points represent the results of theoretical analysis. On the one hand, by comparing the simulation results with the theoretical analysis, it can be observed that the analytical results from the generalized mean field theory align well with the numerical simulation results. On the other hand, for these given values of r, increasing the critical threshold θ_{th} will significantly enhance the level of cooperation in the system. However, there is a diminishing marginal effect on the increase in the level of cooperation. After all, the larger the critical threshold θ_{th}, the fewer agents have the

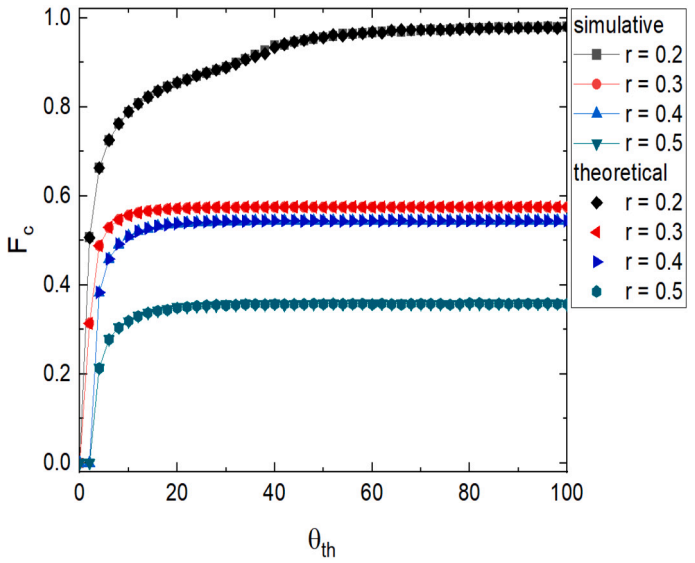


Fig. 2. For some different values of r, the frequency of cooperation F_c as functions of critical threshold θ_{th} at the equilibrium phase. In addition, we made a fitting comparison between the simulation results and the theoretical results. Some of the main parameters are consistent with Fig. 1.

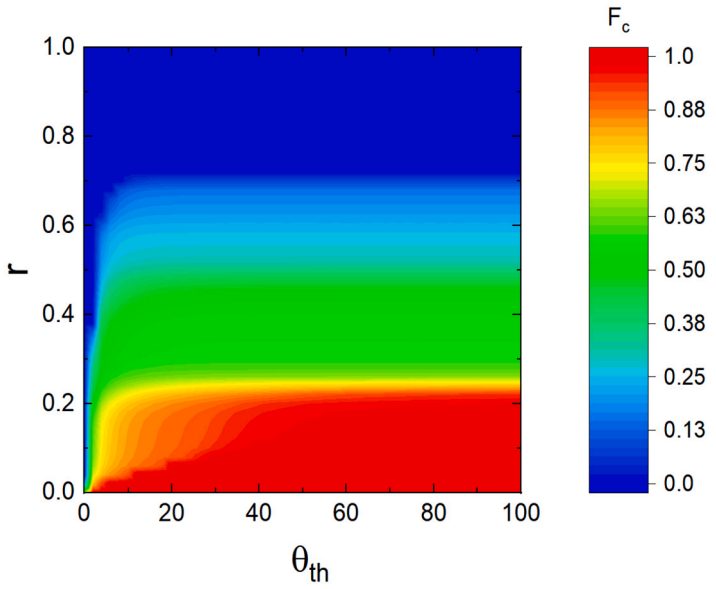


Fig. 3. The $r-\theta_{th}$ parameter plane encoding the frequency of cooperation F_c at the equilibrium stage. The color progression spanning from red through yellow, green, and cyan to blue visually encodes the continuous transition from universal cooperation (the frequency of cooperation $F_c = 1$) to complete defection (the frequency of cooperation $F_c = 0$). Some of the main parameters are consistent with Fig. 1.

opportunity to participate in the prisoner's dilemma game. Correspondingly, when the critical threshold θ_{th} increases to a certain level, almost all agents can only engage in the snowdrift game, and the level of cooperation naturally cannot be further improved. Meanwhile, it can be found that, once the maximum value of strategy stickiness θ_{max} reaches a certain level, its impact on the evolution of cooperation is negligible. Only when the value of θ_{max} is relatively small can it affect the result of the evolution of cooperation. Thus, to draw a more accurate analogy between our strategy stickiness and the natural age, we neither set it too large nor too small. Instead, we have set the maximum value at $\theta_{max} = 100$, as it holds greater practical significance.

To systematically investigate the impact of threshold θ_{th} and r on the evolution of cooperation in a broader scope, we construct the $r-\theta_{th}$ parameter plane by using color gradients to represent varying frequency of cooperation F_c, as shown in Fig. 3. By comparing Figs. 2 and 3, it can be found that, the trends in the evolution of cooperation they exhibit are qualitatively consistent. Upon vertical examination, for constant θ_{th} values, systematic observations reveal that increasing r values correlate with sequential color transitions in the diagram (progressing from red through yellow, green, light blue, to dark blue), thereby indicating that the willingness to cooperate gradually weakens. Naturally, it follows that as the value of r increases, the intensity of the social dilemma escalates,

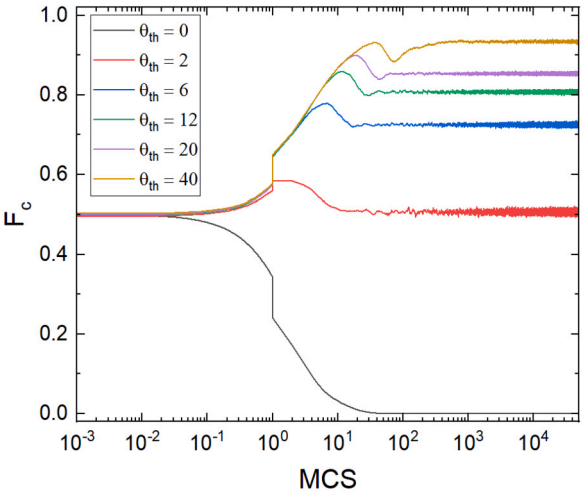


Fig. 4. Under kinds of values of critical threshold θ_{th}, time evolution sequence of the frequency of cooperation F_c when $r = 0.2$. Some of the main parameters are consistent with Fig. 1.

thereby rendering the emergence and diffusion of cooperative strategies progressively more challenging. A cross-sectional analysis reveals that the cooperation rate exhibits a significant positive correlation with the threshold θ_{th}, yet this upward trend proves non-sustained. Furthermore, the influence of critical threshold θ_{th} on the evolution of cooperation markedly displays complex behavior that varies nonlinearly across distinct intervals of r. In general, after $\theta_{th} > 5$, the light blue and green regions seem to change little with the increase of θ_{th}. However, the red region, in the range of $0 < r < 0.2$, exhibits relatively significant variations as θ_{th} increases. More specifically, the larger the value of θ_{th}, the deeper the red area, indicating that a greater number of agents are incorporated into the SDG environment. Nevertheless, when θ_{th} increases to around 50, the red area no longer changes, which means that at this point there are almost no agents participating in the PDG anymore. Combining Figs. 2 and 3, we can infer that moderately adjusting the maximum value θ_{max} upwards or downwards will not fundamentally change the outcome of the evolution of cooperation. However, if the maximum value θ_{max} is set to an extremely small level, a fundamental change in the evolution of cooperation will occur. The above results further confirm that, the threshold θ_{th} constrains the evolution of cooperation in spatial heterogeneous social dilemmas with dynamic self-adaptive matching.

3.2. *Microscopic analysis of the underlying mechanism*

The macro results indicate that, the more agents involved into the SDG environment, the more conducive to cooperation, which are consistent with the intuitionistic analysis. However, we still do not know the exact details of the evolution of cooperation. After all, the heterogeneous social dilemmas created by asymmetry and dynamics has generated many uncertainties. Then, we need to thoroughly clarify this issue from the microscopic perspective.

Thus, analysing the temporal dynamics of cooperation at the microscopic scale contributes to clarify key mechanisms underlying evolution of cooperation in spatial heterogeneous multi-game systems. To this end, for selected critical threshold θ_{th} values with r maintained at 0.2, we present the time series of the frequency of cooperation F_c, as shown in Fig. 4. Our above analysis indicates that, all agents are classified into the adults and compelled to confront the rigorous challenges posed by PDG when $\theta_{th} = 0$. As we can observe in Fig. 1, the cooperators have completely annihilated even at $r = 0.022$. Under current conditions with $r = 0.2$, cooperators are rendered utterly incapable of resisting defectors' onslaught given the amplified social dilemmas. So it can be found that, the frequency of cooperation F_c exhibits a monotonic decrease trajectory, ultimately culminating in cooperation extinction as the system undergoes phase transition into full defection dominance. When $\theta_{th} > 0$, the situation that is not conducive to the formation and maintenance of cooperation will be reversed. The frequency of cooperation F_c demonstrates a monotonic ascent trajectory devoid of the initial defector incursion phases characterized by transient cooperative erosion, which are the phenomena ubiquitously documented in canonical evolutionary frameworks [49]. What is more, these evolutionary trajectories decrease slightly after a short stay at the peak. In other words, these time series all have meta-stable state when $\theta_{th} > 0$. After that, these curves decline and lift slightly, then enter the relatively stable dynamic equilibrium. Furthermore, the larger value of θ_{th}, the higher the frequency of cooperation F_c, and the more obvious the trend of the curves falling first and then rising slightly after reaching the peak. Together, these findings collectively demonstrate that within spatial heterogeneous social dilemmas with dynamic adaptive coupling, the SDG reconfigures the game environment through the exogenous payoff restructuring, making the evolution state extremely complex and difficult to accurately predict.

Here, it is particularly worth highlighting that, to obtain more in-depth microscopic details of the evolution of cooperation, we need to pay attention to the agent's strategy configuration for some key steps during the evolution process. Thus, based on the time series of the frequency of cooperation depicted in Fig. 4, we select several pivotal time steps and generate some snapshots of those specific moments, as shown in Fig. 5. The lines from top to bottom, the values of θ_{th} are 0, 2, 6, 12, 20, and 40, respectively. The columns

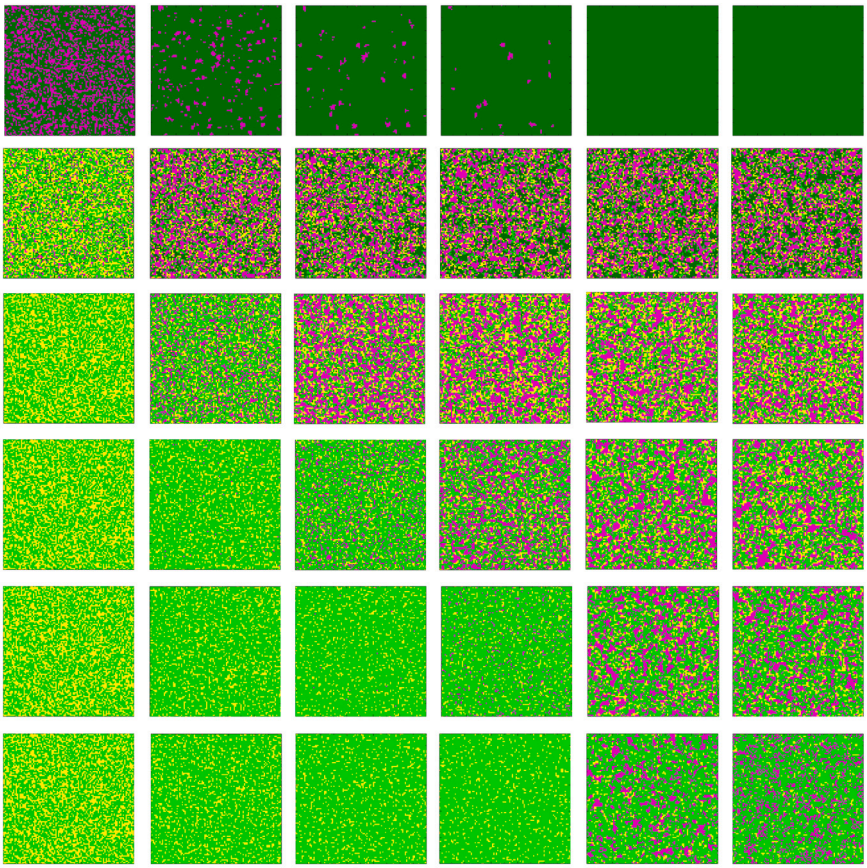


Fig. 5. For several critical thresholds θ_{th}, snapshots of the strategy evolution with some representative time steps when $r = 0.2$. The lines from top to bottom, the values of θ_{th} are 0, 2, 6, 12, 20, and 40, respectively. The columns from left to right, the steps are 0, 5, 10, 15, 100, and 50000, respectively. The magenta and green dots indicate cooperators of adults and juveniles, respectively. The olive-green and yellow spots refer to defectors of adults and juveniles, respectively. Some of the main parameters are consistent with Fig. 1.

from left to right, the steps are 0, 5, 10, 15, 100, and 50000, respectively. The magenta and green dots indicate cooperators of adults and juveniles, respectively. The olive-green and yellow spots refer to defectors of adults and juveniles, respectively. For comparison, we still first look at the strategy evolution snapshots when $\theta_{th} = 0$. On the one hand, since all agents are identified as adults, they only participate in the PDG. On the other hand, the system exhibits acute social dilemmas under $r = 0.2$, thereby preventing cooperators from establishing cohesive clusters amidst sustained onslaught by defectors. Thus, it can be found that, although magenta cooperators try their best to form clusters against the defectors, they disintegrate in the olive sea at last. Ultimately, the system falls into the state of mutual defection. On the contrary, when $\theta_{th} > 0$, the fate of cooperators is radically rewritten. In terms of strategic persistence, there are always some agents who fail to surpass the given threshold θ_{th}, and thus participate in the SDG as juveniles. Even so, it is imperative to emphasize that, the inverse proportionality between threshold magnitude and agents' transition probability into adult categorization not only quantifies boundary permeability but also implicates nonlinear phase transitions in developmental trajectory modeling. For instance, when $\theta_{th} = 2$, we can find that even in the initial stage, there are some agents involving in the PDG, which is caused by the asynchronous strategy update. Over time, magenta and olive-green spots are increasing and peak at steps $= 15$. After that, magenta clusters slightly contract due to the division of the green and yellow ribbons. Therefore, the root cause of the system cooperation peak and then decline is the segmentation of cooperation clusters caused by SDG, which is similar to the report in [19]. When the system enters the stable stage, the four strategies dynamically coexist.

However, if the threshold θ_{th} is continuously increased, we can observe counterintuitive phenomena. On the one hand, the later the adults involved in the PDG emerge. On the other hand, even though some agents exhibit strategic persistence sufficient to surpass the critical threshold θ_{th} and transition into an 'adult' state (*e.g.,* reaching maturity in decision-making capabilities), they can only participate in the PDG as cooperators. This counterintuitive constraint has sparked significant interest in our investigation of the underlying mechanisms. Intuitively speaking, in the SDG, defection is the better choice if the opponent cooperates. Consequently, defectors in the SDG will persist and eventually become embedded within the PDG environment. Surprisingly, we do not find defection in the PDG under the larger threshold conditions, which is counterintuitive. For instance, when $\theta_{th} = 20$, there are only cooperators that participate in PDG, and agents with cooperative and defective strategy take part in SDG. Furthermore, as reported in [19], the unique Nash equilibrium of the SDG prevents cooperators from forming compact clusters, instead driving them to organize into filamentous or strip-like patterns, which limits the further development of cooperation. Here, with the increase of critical threshold

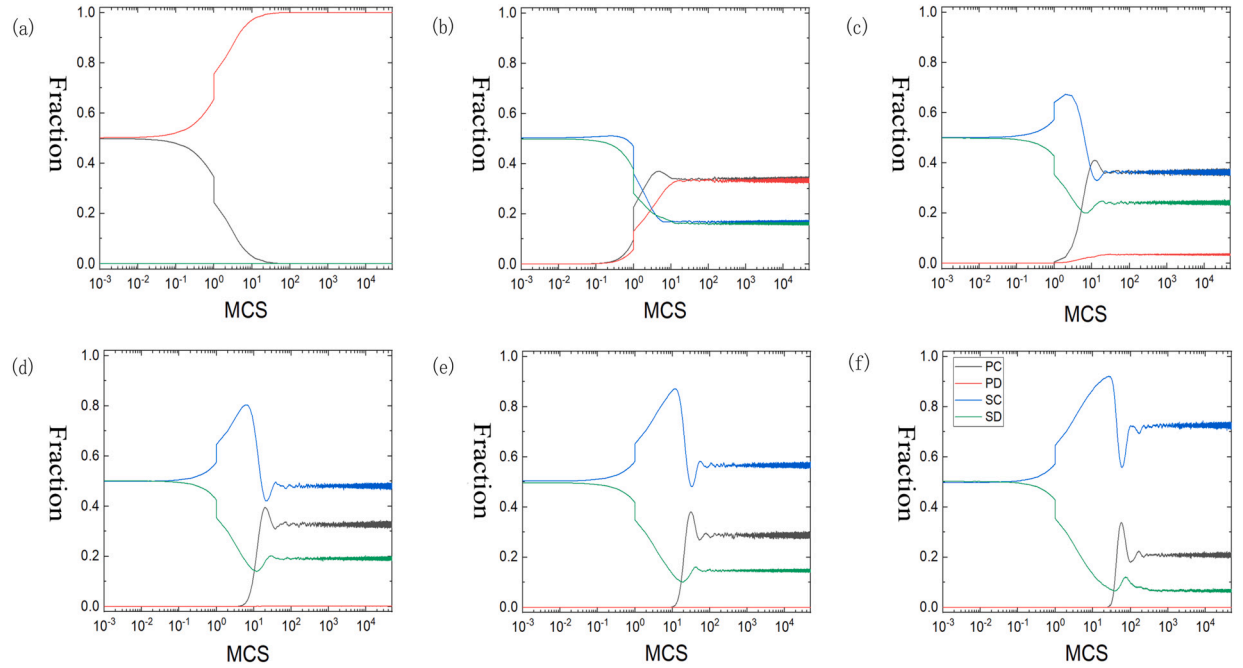


Fig. 6. The time series of different strategies when $r = 0.2$ for several critical thresholds θ_{th}. panels (a) and (f), the values of critical threshold θ_{th} are 0, 2, 6, 12, 20, and 40, respectively. Some of the main parameters are consistent with Fig. 1.

θ_{th}, more and more agents participate in the SDG, which in turn causes cooperators in the PDG environment to become fragmented into filamentous or strip-like patterns. Hence, the larger the value of θ_{th}, the more severely the magenta clusters are segmented.

With the increase of threshold θ_{th}, the phenomena revealed by the snapshots of strategy evolution have caught our attention. Therefore, it is necessary to conduct the comprehensive examination of the evolutionary frequencies of different strategies in each case. Thus, Fig. 6 presents the time series of the fraction of different strategies. The panels from (a) to (f), the critical threshold parameter θ_{th} are 0, 2, 6, 12, 20, and 40, respectively. The black and red curves correspond to cooperators and defectors in PDG, respectively. In addition, the blue and green lines are cooperative and defective strategies in SDG, respectively. Collectively, all these curves show bifurcation phenomena. In addition, different values of θ_{th} make discrete strategy exhibit marked divergence in evolutionary trajectories, which is a manifestation of nonlinear dynamics in strategy selection mechanisms that fundamentally redefines equilibrium attainment pathways. To be specific, when $\theta_{th} = 0$, all agents play PDG with their opponent. At this moment, the red evolutionary trajectory demonstrates strict monotonic convergence towards complete defective strategy fixation, whereas the black counterpart undergoes irreversible divergence leading to cooperation extinction. This diametric opposition in strategic dynamics unveils fundamental mono-stable characteristics. When $\theta_{th} > 0$, all agents participate in the SDG at the initial stage due to our setup. So at the beginning of evolution, the blue and green curves are 50%, respectively. Over time, when $\theta_{th} = 2$, the back and red lines begin to rise as the blue and green curves fall. Eventually, these four curves tend to balance in the dynamic. Because the threshold θ_{th} is relatively low at this time, it is not difficult for agents to overcome the barrier to be adulthood. Therefore, the proportion of agents participating in the PDG is much higher than those participating in the SDG. Notably, both PDG and SDG demonstrate almost the same level of cooperation, while the fractions of defection in PDG and SDG also exhibit parallel evolutionary trajectories. Moreover, the number of cooperators is much higher than that of defectors.

Progressive elevation of the critical threshold θ_{th} triggers behavioral complexities within the system, revealing phase-dependent evolutionary singularities. For instance, when $\theta_{th} = 6$, the time series of cooperators involved in the PDG and the SDG both experience the process of first rising to meta-stable state, and then falling to dynamic stability. For defectors, the green curve rises slightly after descending to the bottom and enters the dynamic stable phase at last; while the red line only climbs slightly, and then initiates the dynamic stability. At the same time, it can be found that, with the help of agents who take part in SDG, this multi-level selection framework demonstrates not only enhancement in global cooperation through non-equilibrium phase transitions, but more significantly reveals cross-model synergistic effects. To be more specific, cooperation prevalence in the PDG exhibits nonlinear amplification when coupled with SDG mediated evolutionary pathways, which fundamentally changes traditional single-game equilibrium predictions while establishing new attractor basin bifurcation phenomena in cross-strategy phase space. Upon progressively increasing the threshold θ_{th}, we can observe the emergence of even more pronounced nonlinear characteristics. Panels from (d) to (f), the red line remains at 0 without any upward momentum. What is more, in the initial phase the black curve follows the blue curve in a continuous ascent, and both subsequently exhibit varying degrees of decline after reaching their peaks. Moreover, in these panels the blue curve reaches its peak before the black one, and the peak of the black line nearly coincides with the trough of the blue curve. In particular, the larger the value of critical threshold θ_{th}, the higher the peak of both the blue and black curves, and the lower the trough of them.

The separation amplitude between the red and black curves exhibit a monotonic increase with higher θ_{th} values. Combined with Figs. 5 and 6, we now answer why, for larger value of θ_{th}, defectors involved in the PDG will no longer emerge. The higher value of θ_{th} makes it more difficult for agents to overcome barriers and join the adult group. Consequently, all agent will engage in the SDG environment at the initial stage. The SDG framework exhibits bistable strict Nash equilibria, characterized by asymmetric strategy profiles (D,C) or (C,D) [19]. In other words, whatever the strategy of the opponent adopts, cooperation is better than defection. Cooperation persists as the dominant strategy even when confronted with adversarial defection, as empirical game-theoretic analyses demonstrate collaborative reciprocity yields benefits exponentially surpassing its implementation costs: a Pareto-superior outcome starkly contrasting with the zero-sum destitution endemic to mutual non-cooperation. Therefore, for larger value of θ_{th}, cooperative agents will be retained and given the opportunity to participate in PDG, while defective agents can by no means overcome the barrier. The phenomenon depicted in Fig. 6 imparts a valuable lesson: everything has its dual aspects. Even when we are in the quite dire or unfavorable situation, we should adjust our mindset and confront it with a positive attitude.

Through multi-angle and multi-level macroscopic and microscopic analyses, within the context of spatial heterogeneity in social dilemmas, if dynamically adaptive matching based on strategic stickiness is implemented, we can observe more intriguing phenomena in the evolution of cooperation. Specifically, when the threshold values θ_{th} is quite large, there are no agents adopting the defection strategy to engage in the prisoner's dilemma game. Notably, our research, which focuses on strategic stickiness, shares many similarities with previous study on stickiness of social ties [60]. In this model [60], even after random migration, individuals are permitted to maintain some connections with their former partners, which better aligns with real-world scenarios. Traditional perspectives suggest that random migration typically disrupts cooperation because it disturbs the clustering effect of cooperative groups. However, the findings of this study indicate that, when a certain number of social connections are maintained, even completely random migration can facilitate the emergence of cooperative behavior [60]. Therefore, constructing models that better capture the complex and dynamic nature of real-world scenarios yields research findings of profound significance for our understanding of the emergence and sustainability of cooperation.

3.3. Robustness analysis and verification

It is well-established that degree distributions of real-world network typically follows either Poisson distribution or power-law pattern. Concurrently, some networks may exhibit small-world characteristics featuring high clustering with short path lengths. Apparently, lattice network is inherently incapable of replicating the above describing topological characteristics, which naturally provokes a serious question: the scientific validity and generalizability of corresponding research findings become fundamentally questionable when such idealized models demonstrably lack essential real-world topological signatures. Meanwhile, the strategy update mode will also profoundly influence the trend and law of the evolution cooperation. To solve the above issues, we will explore the robustness and resilience of the model through other network configurations and strategy update patterns.

To decisively eliminate these methodological doubts, we first systematically select some network architectures with other normative characteristics as agent-interaction topologies. Subsequently, we launch a series of controlled comparative experiments to rigorously examine topological effects on evolutionary dynamics, as shown in Fig. 7. Panels from (a) to (f), the results are obtained on random regular network, Erdös-Rényi random network, regular small world network, Watts-Strogatz small-world network, Barabási-Albert scale-free network, and square lattice network with Moore neighborhoods, respectively. In addition, the scale of all the analyzed networks remains $N = 10,000$ and the average degree keeps $< k >= 4$, except for the square lattice network with the Moore neighborhood. The square lattice network with Moore neighborhood is the lattice configuration where each node interacts with eight nearest neighbors (including diagonal connections), resulting in the distinctive $< k >= 8$ topology. This deliberate parameterization ensures topological comparability while systematically investigating degree distribution effects on evolutionary outcomes. In general, the results demonstrated in Fig. 7 maintain qualitative consistency with those presented in Fig. 1. More specifically, these results reveal the monotonic positive correlation where the level of cooperation escalates with increasing threshold θ_{th} values. While concurrently, the extinction of cooperation r_c exhibits proportional scaling with the primary threshold parameter across all tested conditions. This coupling phenomena suggest the fundamental trade-off mechanism between cooperative resilience and environmental selection pressure.

Furthermore, the unique properties of the network itself also have a significant impact on the outcome of the evolution of cooperation. For instance, random regular network and regular small world network have exactly the same degree distribution with $< k >= 4$. By comparing the results in panels (a) and (c), we can find there are two distinct differences for all threshold θ_{th} values. On the one hand, for the range of r that maintains full cooperation, regular small world network is obviously better than that of random regular network. On the other hand, for the threshold r_c where cooperation disappears completely, random regular network is significantly better than that of regular small world. The aforementioned phenomena are quite different from the conclusion of Hauert and Szabó [50], that is, replacing some local connections with long-range edges always provides favorable opportunities for cooperators. In fact, it makes sense. After all, the single-configuration PDG or SDG social dilemmas in the system have been replaced by heterogeneous social dilemmas with adaptive matching, which not only increases the complexity of agent identification social dilemmas but also creates favorable conditions for the evolution of cooperation. However, the double-edged sword effect exhibited by long-range edges is also extremely pronounced. When the intensity of the social dilemma is low, a small number of long-range edges compel agents to seek assistance from neighboring peers, forming compact clusters that enhance the level of cooperation, potentially even achieving full cooperation. In contrast, under stronger social dilemmas, abundant long-range edges enable cooperative agents to seek partnerships over greater distances, thereby significantly raising the threshold r_c for the extinction of cooperation.

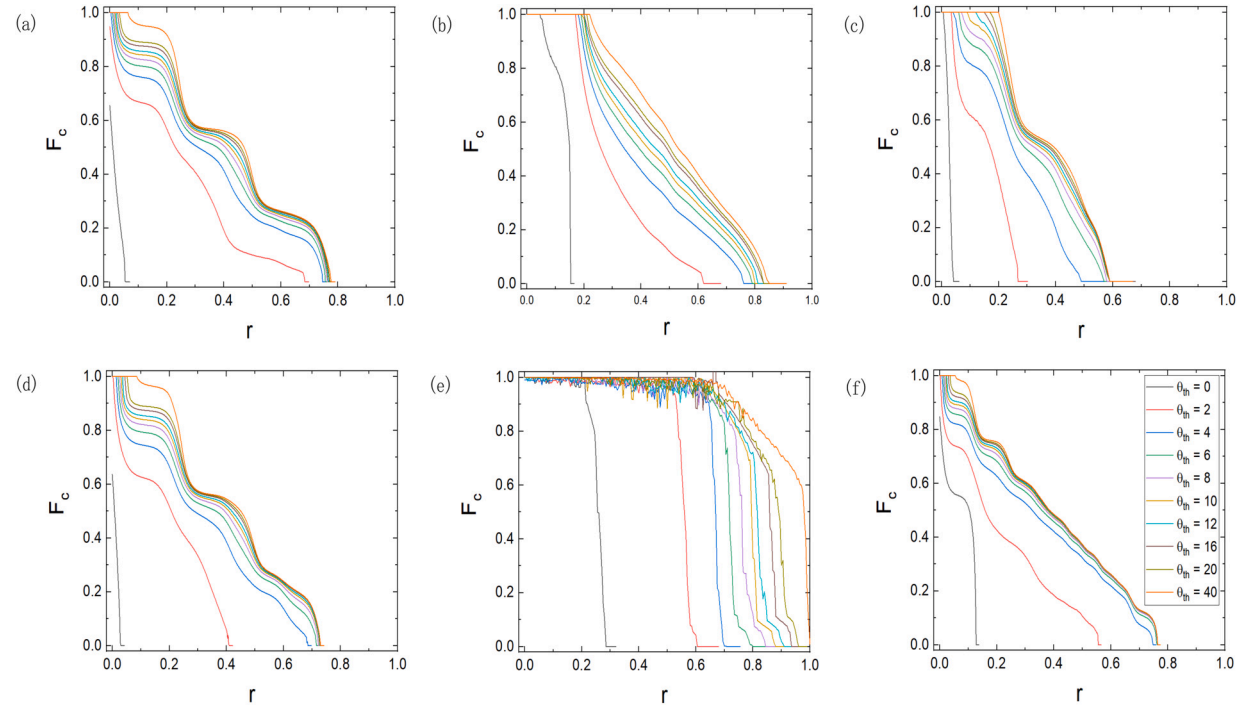


Fig. 7. For different network structures, he frequency of cooperation F_c as functions of cost-to-benefit radio r at the equilibrium phase. Panels from (a) to (f), the results are obtained on random regular network, Erdös-Rényi random network, regular small world network, Watts-Strogatz small-world network, Barabási-Albert scale-free network, and lattice network with Moore neighborhoods, respectively. Except for lattice network with Moore neighborhoods having an average degree of $< k >= 8$, all other networks exhibit the average degree of $< k >= 4$. Some of the main parameters are $MCS = 5 \times 10^4$ and $K = 0.1$.

Moreover, comparing the results in panels (b) and (e), Erdös-Rényi random network and Watts-Strogatz small-world network just exhibit phenomena that are exactly opposite to those observed previously. For Erdös-Rényi random network and Watts-Strogatz small-world network, their paths are all relatively short. However, the clustering coefficient of Watts-Strogatz small-world network is much higher than that of Erdös-Rényi random network. The high clustering coefficient induces an echo chamber effect [58,59], making it difficult for agents to adapt to dynamic environments and hindering the spread of cooperation at last. Panel (e) exhibits the most pronounced cooperative behavior when compared to all other panels analyzed, as demonstrated by the results. However, we can observe large fluctuations, which are caused by the power-law distribution of scale-free networks. In particular, we also observe surprising phenomena in panel (f). According to previous studies [61], the level of cooperation can be significantly enhanced in a single PDG once the neighborhood is appropriately enlarged, which is indeed verified when $\theta_{th} = 0$. While $\theta_{th} > 0$, the outcomes of cooperation are not always better than those shown in Fig. 1. In detail, when the intensity of social dilemmas is relatively weak, the interval of r sustaining full cooperation narrows; whereas for the strong social dilemmas, the critical threshold r_c requires to sustain the evolution of cooperation has demonstrated a significant increase. The above results indicate that, even if the characteristics of the networks themselves have some certain impact on the evolution of cooperation, the qualitative positive correlation between the level of cooperation and the threshold θ_{th} is almost the same. More specifically, Fig. 7 validates that our model has a certain degree of robustness and resilience against kinds of network structures in terms of the evolution of cooperation.

At last, we further test the robustness and elasticity of the model on the evolution of cooperation from the perspective of strategy update pattern. To this end, Fig. 8 plots the complete $r - \theta_{th}$ parameter plane, encoding the variation of the frequency of cooperation F_c under synchronous strategy update pattern. Comparative analysis of the results from Figs. 8 and 3 demonstrates that, the evolution of cooperation, on the whole, qualitatively exhibit the strong degree of consistency. Nevertheless, certain nuances persist that warrant closer scrutiny within the experimental framework. Intuitively speaking, the red zone in Fig. 8 is larger than that in Fig. 3. These phenomena suggest that in our model, the synchronous strategy update pattern is more conducive to the evolution of cooperation, which is completely contrary to previous studies [57]. Empirical evidence consistently demonstrates that asynchronous strategy update pattern confer distinct evolutionary advantages in fostering the evolution of cooperation, with rigorous validation across both prisoner's dilemma game and public goods game frameworks [62,63]. While for the snowdrift game, synchronous strategy update pattern allows for a higher level of cooperation [64]. Thus, whether the efficacy of asynchronous or synchronous strategy update protocol is more conducive to the evolution of cooperation fundamentally depends on the inherent type of social dilemma in the system. In terms of this model, there is a monotonic positive correlation between the threshold θ_{th} and the agents' participation in the SDG. The larger the value of θ_{th}, the more agents are involved into the SDG environment. Finally, we can find the higher level of cooperation in the case of synchronous strategy update mode.

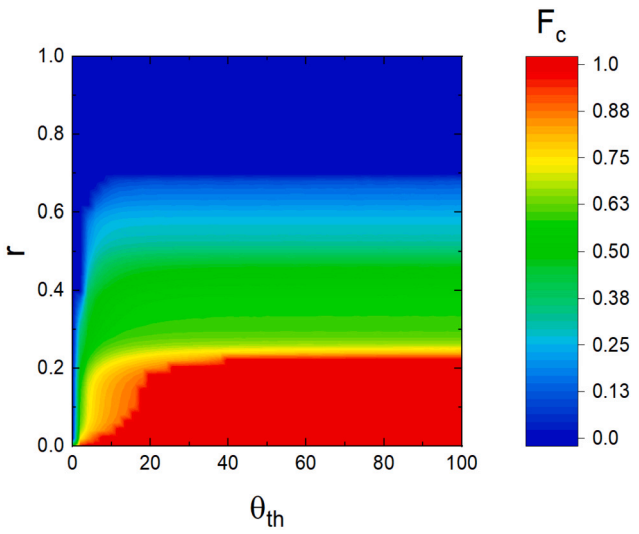


Fig. 8. For the synchronous strategy update pattern and lattice network topology with Von Neumann neighborhood, the $r-\theta_{th}$ parameter plane encoding the frequency of cooperation F_c at the equilibrium stage. All other parameters remain identical to those in Fig. 3.

4. Conclusions

To put it simply, we have constructed a heterogeneous social dilemmas game model incorporating dynamic adaptive matching mechanisms, designing to systematically investigate both macro-level emergent patterns and micro-level behavioral regularities in the evolutionary dynamics of cooperation, as well as the robustness and resilience of cooperative maintenance and propagation.

In detail, each agent is endowed with an additional attribute beyond their strategy, defined as strategy stickiness or persistence, and the initial value is set to be 0. To adequately characterize the natural attribute of strategy persistence in agents, we draw an analogy between it and age-related characteristics. More specifically, the agent's strategic persistence bears some resemblance to natural age-related characteristics. To be specific, if an agent's strategy remains unchanged during the strategy update process, its strategy persistence increases one unit; otherwise, the strategy stickiness is reset to the initial value of 0, signifying the deactivation of the existing agent and the generation of a new one. In addition, to better achieve closer alignment with biological life-cycle analogies, we impose a persistence cap constraint: when the agent's strategy stickiness accumulates to 100, this automatically triggers termination of the incumbent agent and spawning of a successor one through deterministic replacement protocols. Moreover, the strategy stickiness of the newly generated agent will be set to the initial value of 0. Then, we further institute a critical threshold θ_{th}, implementing developmental stage partitioning through comparative evaluation of agent $x's$ current strategy stickiness against this benchmark. This mechanism bifurcates the population into juvenile population ($\theta_x < \theta_{th}$) and adult cohort ($\theta_x \geq \theta_{th}$). As protected entities, the juvenile group will participate in the snowdrift game with relatively mild social dilemmas, whereas the adult cohort, functioning as primary obligation bearers, must engage in high-stakes prisoner's dilemma game with amplified social conflict structures. In particular, we acknowledge that the factors influencing an agent or entity's adjustment of strategies or participation in types of social dilemmas are diverse and multifaceted. However, to highlight the dynamic perception capabilities of agents in social dilemmas, we have simplified the abstraction of real-world scenarios. Despite this simplification, we have captured intriguing experimental phenomena that hold theoretical and applied value.

Through a large number of Monte Carlo simulations, we macroscopically reveal that, cooperation prevalence exhibits monotonic enhancement with the increase of critical threshold θ_{th}. This emergent pattern stems fundamentally from the density-dependent snowdrift game participation effect: the subpopulation engages in snowdrift game displays a statistically significant positive correlation with θ_{th}. However, this enhancement effect presents marginal diminishing returns, fundamentally constrained by the saturation boundary of snowdrift game dominance. When all agents become snowdrift game participants, the evolution of cooperation reaches saturation. In particular, we compare the theoretical results derived from the generalized mean-field approach with those from Monte Carlo simulations, and find that they can fit well with each other. Taking a step further, we explore the dynamics of the evolution of cooperation through both strategy evolution snapshots and temporal sequences, and find surprising phenomena in the process. With increasing θ_{th} values, the frequency of cooperation in prisoner's dilemma game and snowdrift game will exhibit successive growth phases, peaking at differential thresholds before undergoing distinct decline patterns ultimately converging towards dynamic equilibrium. Notably, the frequency of cooperation trajectories present near-perfect phase locking between snowdrift game troughs and prisoner's dilemma game peaks, revealing counterintuitive synchronization across dilemma domains. In stark contrast, the frequency of defection in the snowdrift game shows a continuous decline and stabilizes in the shock. Contrary to the theoretical prediction of intuition, for quite larger value of θ_{th}, defectors in the prisoner's dilemma game consistently fail to gain evolutionary prevalence, presenting a paradoxical deviation from established behavioral norms. This counterintuitive phenomenon suggests that even in harsh social dilemmas, cooperation is still the order of the day.

At last, we test the evolutionary robustness and dynamical resilience of mechanisms in our model through other common network configurations and synchronous strategy update protocol. For other network configurations, although the intrinsic structural properties of alternative network topologies impart modulations on the dynamics of cooperation, the evolutionary trajectories maintain qualitative consistency with core findings across the entire threshold parameter space, indicating that our model is relatively resilient to the network structure in terms of the evolution of cooperation. As for other strategy update pattern, the evolution of cooperation under synchronous strategy update condition is obviously better than that under asynchronous strategy update mode due to the existence of snowdrift game. The potential reason for this phenomenon is that under synchronous strategy updating protocol, the inherent incentive structure of snowdrift game leads to enhanced stage selection of cooperative dynamics, whereas asynchronous strategy updating regime shows a characteristic frustration effect that inhibits cooperative meta-stability.

In summary, by incorporating strategy viscosity into the study of the evolution of cooperation, we establish a game model for heterogeneous social dilemmas with dynamic adaptive matching. Despite the limitations of our simplified setup, which prevents it from precisely depicting and describing the highly complex scenarios in the real world, the results we obtained are nonetheless intriguing. Specifically, cooperation aligns more closely with the expectations of selfish individuals. This formalism unravels cooperation emergence mechanisms in high-dimensional non-equilibrium asymmetric parameter spaces, which helps us to understand cooperation stabilization pathways in real scenarios. Moving forward, drawing on the research approaches of previous study [43], conducting research on the evolution of cooperation under adaptive matching mechanisms tailored to different intensities of the same type of social dilemma is undoubtedly a research direction worthy of significant attention. Meanwhile, we can also refer to [44] to investigate the evolution of cooperation when agents dynamically switch between pairwise and group interactions based on actual circumstances. Of course, it will be of greater significance to decode cooperation emergence in networks with non-stationary asymmetric dynamics through empirical behavioral data-sets, which will be critical to accurately regulate and optimize the trajectory of the evolution of cooperation. We hope that our research can provide useful reference for understanding the evolution of cooperation in high-dimensional non-equilibrium asymmetric regimes.

CRediT authorship contribution statement

Xiaopeng Li: Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing. **Xuexue Yang:** Data curation, Formal analysis, Investigation, Software, Writing – original draft. **Xiuli Zhang:** Data curation, Formal analysis, Investigation, Software, Writing – original draft. **Weiwei Han:** Methodology, Software, Validation, Visualization, Writing – review & editing. **Yan Xu:** Methodology, Visualization, Writing – review & editing.

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Data availability

Data will be made available on request.

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